INCH-POUND

FF-S-107D
14 JANUARY 2022
SUPERSEDING
FF-S-107C
27 MAY 1969

DETAIL SPECIFICATION SHEET

SCREW, TAPPING AND DRIVE

Inactive for new design after 15 December 2000
and is no longer used, except for replacement purposes.

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

Comments, suggestions, or questions on this document should be addressed to DLA Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or emailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5305

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1. Scope. This specification covers the requirements for thread cutting, thread forming and thread rolling tapping screws, and metallic drive screws.

1.2. Classification.

1.2.1. Types. Tapping and drive screws shall be the following types, as specified (see 6.2).

Type A - Thread forming, gimlet point.
Type AB - Thread forming, gimlet point.
Type B - Thread forming, blunt point.
Type BP - Thread forming, cone point.
Type C - Thread forming, blunt point.
Type D - Thread cutting, blunt point.
Type F - Thread cutting, blunt point.
Type G - Thread cutting, blunt point.
Type T - Thread cutting, blunt point.
Type BF - Thread cutting, blunt point.
Type BT - Thread cutting, blunt point.
Type SF - Thread rolling, high performance, blunt point.
Type SW - Thread rolling, high performance, blunt point.
Type TT - Thread rolling, high performance, blunt point.
Type U - Drive screw, pilot point.

Note: Type BG – thread cutting, blunt point has been removed from this specification. Use type BF or BT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

1.2.2. Head styles. Tapping and drive screws shall be of the following head styles:

	Head	Slotted	Recessed	Plain
*	Flat	-	X	-
**	Flat	X	X	-
**	Undercut flat	X	X	-
**	Flat trim	-	X	-
**	Oval	X	X	-
**	Undercut oval	X	X	-
**	Oval trim	-	X	-
	Pan	X	X	-
	Fillister	X	X	-
	Truss	X	X	-
***	Hexagon	X	X	-
***	Hexagon washer	X	-	X
	Drive screw	-	-	X
	Round	X	-	X

* 99˚ – 101˚ Bearing surface
** 80˚ – 82˚ Bearing surface
*** Regular or indented, as specified (see 6.2)

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government Documents.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 – DOD Preferred Methods for Acceptance of Product.

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3. Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ).

ASQ H1331 – Zero Acceptance Sampling Plans

(Copies of this document are available for purchase online from www.asq.org.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

INDUSTRIAL FASTINERS INSTITUTE (IFI) STANDARD.

IFI-112 – High Performance Thread Rolling Screws.

(Copies of this document are available for purchase online from www.indfast.org).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL.

SAE AIR4127 – Steel: Chemical Composition and Hardenability

SAE AMS2700 – Passivation of Corrosion Resistant Steels

SAE AMS-QQ-P-416 – (R) Plating, Cadmium (Electrodeposited)

(Copies of this document are available for purchase online from www.sae.org).

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME) STANDARD.

ASME B18.6.3 – Slotted and Recessed Head Tapping Screws and Metallic Drive Screws.

(Copies of this document are available for purchase online from www.asme.org).

2.4. Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

3. REQUIERMENTS

3.1. Materials.

3.1.1. Carbon steel. Unless otherwise specified in the contract or order, screws shall be manufactured from carbon steel (i.e., AISI 1016, 1018, 1019, 2021, 1022 or 1024). Carbon steel screws shall be heat treated or case hardened to meet all the requirements specified herein.

3.1.2. Corrosion – resisting steel. When specified, corrosion-resisting steel shall be UNS S4100 (hardened) in accordance with SAE AIR4127. CRES screws shall meet all the requirements specified herein, except for the performance requirements of 3.5.1 (see 3.5.2).

3.1.3. Other Material. When other material is used for the manufacturer of screws, such material shall conform to the material specification referenced in the contract or order. These screws shall meet all the requirements specified herein, except for the performance requirements of 3.5.1 (see 3.5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.2. Dimension and tolerances. Dimension and tolerance shall be in accordance with ASME B18.6.3, except for types SF, SW and TT which shall be in accordance with IFI-112.

3.3. Protecting coating.

3.3.1. Unless otherwise specified (see 6.2), screws shall have a cadmium plating on carbon and alloy steel and a passivation treatment for corrosion-resisting steel.

3.3.1.1. Cadmium plating. Cadmium plating shall be Type II, Class 3 in accordance with AMS-QQ-P-416.

3.3.1.2. Passivation. Passivation treatment shall be in accordance with SAE AMS2700.

3.3.2. Hydrogen embrittlement. Cadmium plated screws shall be subjected to an embrittlement relief treatment in accordance with AMS-QQ-P-416, as soon as possible after plating, to minimize the resulting embrittlement.

3.4. Recessed drive. Cross-recessed drives shall be either Type I or IA in accordance with ASME B18.6.3, as specified (see 6.2). Military use shall be limited to Type I.

3.5. Performance requirements.

3.5.1. Carbon steel tapping screw. Carbon steel tapping screws shall be capable of meeting the drive requirements (all types except BF, BT, SF, SW and TT) and the tortional strength requirements (all types except SF, SW and TT) specified below. Types SF, SW and TT carbon steel tapping screws shall be capable of meeting the mechanical and performance requirements specified in IFI-112.

3.5.1.1. Drive.

a. Thread forming screws shall, without deformation of their own threads, form a mating thread in the test plate until the tapered threads of the point are completely through the test plate.

b. Thread cutting screws shall, without deformation of their own threads, cut or form mating thread in the test plate until the tapered threads of the point are completely through the test plate.

3.5.1.2. Tortional strength. The tortional strength required to cause a failure of carbon steel tapping screws shall equal or exceed the minimum values given in the applicable tables of ASME B18.6.3.

3.5.2. Non-carbon steel tapping screws. The performance requirements of tapping screws made of material other than carbon steel shall be subject to agreement between the manufacturer and the user.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.5.3. Drive screws. Drive screws shall, without shearing of the threads on the screw or breaking the screw, produce mating threads in the test plate.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contractor purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2. Inspection provisions.

4.2.1. Lot. Unless otherwise specified, a lot shall consist of screws of the same type, material, finish, size, and head style produced under essentially like conditions and offered for acceptance at any one time.

4.2.2. Sampling.

4.2.2.1. Sampling for examination. Unless otherwise specified in thew contract (see 6.2), for sampling for examination, an “accept at zero” (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. Acceptable verification levels for MIL-STD-1916 and index values for ASQ H1331 shall be specified in Table I.

4.2.2.2. Sampling for performance test. Unless otherwise specified in thew contract (see 6.2), for sampling for performance testing, an “accept at zero” (c=0) sampling plan such as MIL-STD-1916 at verification level III or ASQ H1331 at an index value of 2.5 shall apply.

4.2.2.3. Sampling for passivation test. A random sample of three screws shall be taken from each lot.

4.2.3. Certification.

4.2.3.1. Material. When specified in the contract or order (see 6.2), the contractor shall furnish a certificate of the chemical composition and physical properties of the material being used (see 3.1.3).

4.2.3.2. Protective coating. When specified in the contract or order (see 6.2), the contractor shall furnish a certificate that the protective coatings (see 3.3) are in accordance with the applicable specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

4.3. Examination.

4.3.1. Visual and dimensional. Each screw taken as specified in 4.2.2.1 shall be examined to determine conformance with this specification. Examination shall be conducted in accordance with Table I. Any screw in the sample containing one or more defects shall be rejected, and if the number of defective screws in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE I. Classification of Defects

Categories	Defects	Inspection Method
Critical	None defined	
Major	Verification level III, Index Value = 2.5	
101	Protrusion gaging of flat head screws (6.4)	Gage
102	Recess penetration gaging (6.4)	Gage
103	Type not as specified	Visual
Minor	Verification level I, Index Value = 6.5	
201	Thread length not as specified	SIE*
202	Screw length not as specified	SIE
203	Body diameter not as specified	SIE
204	Eccentricity not as specified	SIE
205	Bearing surface not as specified	SIE
206	Underhead fillet not as specified	SIE
207	Other dimensions not as specified	SIE
208	Protective coating missing or incomplete	Visual
209	Evidence of rust, pitting or spotting after passivation test (4.4.2)	Visual

*SIE Standard Inspection Equipment

4.3.2. Inspection gages. The contractor's inspection gauges shall be an approved precision type, calibrated to standards traceable to the National Bureau of Standards. Current records of calibration shall be retained.

4.4. Tests.

4.4.1. Performance (All types except SF, SW and TT).

4.4.1.1. Drive and torsional strength (tapping screws). Each tapping screw taken as specified in 4.2.2.2 shall be subjected to a drive and tortional strength test to determine conformance to 3.5.1. The tests shall be conducted in accordance with ASME B18.6.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

4.4.1.2. Drive screw tests. Each drive screw taken as specified in 4.2.2.2 shall be subjected to a drive test to determine conformance to 3.5.3. The test shall be conducted in accordance with ASME B18.6.3.

4.4.2. Passivation. Each screw taken as specified in 4.2.2.3 shall be subjected to a passivation test conducted in accordance SAE AMS2700, and examined as specified in Table I.

4.4.3. Hydrogen embrittlement. Cadmium plated screws shall be tested for embrittlement relief in accordance with AMS-QQ-P-416, except that the screw shall be held under load for 23 hours, and then examined for cracks or fracture.

4.4.4. Mechanical and performance. (Types SF, SW and TT). Each tapping screw taken as specified in 4.2.2.2 shall be subjected to mechanical and performance tests to determine conformance to 3.5.1. The tests shall be conducted in accordance with IFI-112.

5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Intended use.

6.1.1. Thread forming screws. Thread forming tapping screws are intended for use in material where large internal stresses are permissible or desirable to increase resistance to loosening.

6.1.2. Thread cutting screws. Thread cutting tapping screws are intended for use in material where destructive internal stresses are not desirable or where excessive driving torques are encountered.

6.1.3. Drive screws. Drive screws are forced into work by pressure and are intended for making permanent fastenings.

6.1.4. Thread rolling screws. Thread rolling tapping screws are intended for use where performance capabilities beyond those normally expected of other types of tapping screws are required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, head style, drive, size and length of screw.
c. Type of hexagon head (regular or indented) (1.2.2).
d. Material, if required (3.1)
e. Material certificate, if required (4.2.3.1).
f. Protective coating (3.3).
g. Protective coating certificate, if required (4.2.3.2).
h. Cross-recessed type, if required (3.4).
i. Levels of preservation, packaging and packing required (5.1).
j. Marking, if required (5.1).

6.3. Method of inspection. When the inspection procedures specified in section 4 of the specification are not applicable, maximum use of certification or a minimum degree of inspection will be employed as specified in section XIV of the Armed Services Procurement Regulation in which the procurement document should specify the amount and location of inspection and the method for contractor's assurance that the requirements of this specification have been met.

6.4. Gaging and formulas. For information about gaging (protrusion and penetration), approximate hole sizes for various materials formulas refer to ASME B18.6.3.

6.5. Supersession data. This specification includes the requirements of MIL-S-6033, dated 24 March 1950.

6.6. Subject term (key word) listing. Cadmium

6.7. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue."

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AR
Air Force - 99
DLA – GS

Preparing Activity:
DLA – GS7

(Project 5305-2022-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.